

Q.30 Write about dimensionless numbers and its significance.

Q.31 Derive the Bernoulli's theorem.

Q.32 Discuss about colour coding used in industry.

Q.33 Discuss about reciprocating Pump.

Q.34 Write about the compressors.

Q.35 Differentiate between Laminar and turbulent flow with diagram?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Describe about the complete derivation of Venturimeter with diagram.

Q.37 Explain construction and working of Reciprocating Pump with neat and clean Diagram.

Q.38 Write short note on any of three:

- 1) manometer
- 2) specific speed
- 3) Dimensional number
- 4) blowers
- 5) pitot tube

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**3rd Sem / Chem, P&P, Chem Engg. (Spl. Paint Tech.),
Chem Engg. (Spl. Polymer Engg.)**

Subject:-Fluid Flow

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Poise is unit of

- a) Viscosity
- b) Density
- c) Specific gravity
- d) Surface

Q.2 Mercury is used in manometer because of its

- a) High density
- b) very low vapour pressure
- c) Both A & B
- d) None of above

Q.3 The fluid which cannot be compressed and have no viscosity

- a) ideal fluid
- b) Real Fluid
- c) Newtonian Fluid
- d) None of above

Q.4 The Reynold number range is $Re < 4000$ for

- a) Laminar flow
- b) Transitional flow
- c) Turbulent flow
- d) All of above

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- Q.5 _____ takes the form of its container.
 a) Solid b) material
 c) Sand d) Fluid
- Q.6 The flow rate of fluid is measure in
 a) m/s b) m³/s
 c) m²/s d) m/s-2
- Q.7 Before the pump started, we should do is
 a) NPSH b) Cavitation
 c) Priming d) All of above
- Q.8 Bernoulli's equation applications are
 a) venturimeter b) orificement
 c) pitotube d) all of above
- Q.9 Other name of laminar flow is
 a) Streamline flow b) turbulent flow
 c) both A & B d) None of above
- Q.10 Principle of conservation of mass is called
 a) Principle of continuity Equation
 b) Energy law
 c) Pascal Law
 d) None of above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write formula for specific gravity.

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- Q.12 What is Pump?
- Q.13 Write Hydrostatic Law?
- Q.14 What is unsteady Flow?
- Q.15 What is Valve?
- Q.16 Write formula of ideal gas law.
- Q.17 Name any two Pumps used in industry.
- Q.18 Define Fluid?
- Q.19 Write one application of blower.
- Q.20 What is incompressible Flow?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Difference between laminar and turbulent flow?
- Q.22 Write about properties of flow?
- Q.23 Discuss about valves and its type.
- Q.24 Write about cavitation.
- Q.25 Write about friction factor chart.
- Q.26 Write the equation for Pascal law.
- Q.27 Explain Friction Losses in Pipes.
- Q.28 Difference between uniform flow and non-uniform Flow?
- Q.29 Draw the labeled diagram for Rotameter?

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